THE STALK OF THE SHEAF OF HOMOMORPHISMS COMPUTES THE HOMOMORPHISMS OF THE STALKS FOR FINITELY PRESENTED MODULES

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Allowed standard prerequisites (editorial).

- The exactness of the sheaf of homomorphisms in the first variable for a right exact sequence, and in the second variable for a left exact sequence. Author lemma lemma-internal-hom-exact, whose statement is reproduced verbatim as a context span of this card (source lines 3775-3799). Produces the exact sequence whose stalk is compared with the corresponding sequence of the stalks.
- The description of the stalk of the sheaf of homomorphisms as classes of local homomorphisms, and local generation of a module of finite type by finitely many sections. Used inside the counted proof at source lines 3851-3863. Reduces injectivity to testing the given local homomorphism on finitely many generators.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$, $\mathcal{G}$ be $\mathcal{O}_X$-modules. If $\mathcal{F}$ is of finite type then the canonical map*

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})_x \to \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$$

is injective. If $\mathcal{F}$ is finitely presented, this canonical morphism is an isomorphism.

Lemma 0.2. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$, $\mathcal{G}$ be $\mathcal{O}_X$-modules.*

(1) *If $\mathcal{F}_2 \to \mathcal{F}_1 \to \mathcal{F} \to 0$ is an exact sequence of $\mathcal{O}_X$-modules, then*

$$0 \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}_1, \mathcal{G}) \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}_2, \mathcal{G})$$

is exact.

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(2) *If* $0 \to \mathcal{G} \to \mathcal{G}_1 \to \mathcal{G}_2$ *is an exact sequence of* $\mathcal{O}_X$*-modules, then*

$$0 \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}_1) \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}_2)$$

is exact.

Proof. The map sends the equivalence class of (U, φ) in $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})_x$, where $x \in U \subset X$ is open and $\varphi \in \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U)$, to the induced map on stalks at x, namely $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$.

Suppose $\mathcal{F}$ is of finite type. Pick a representative (U, φ) of an element σ in the kernel of the map, i.e., $\varphi_x = 0$. Shrinking U if necessary, choose sections $s^1, \ldots, s^n \in \mathcal{F}(U)$ generating $\mathcal{F}|_U$. Since $\varphi_x(s^i_x) = 0$ and we are dealing with a finite number of sections, we can find an open neighborhood $V \subset U$ of x such that $\varphi_V(s^i|_V) = 0$ for all $i = 1, \ldots, n$. Since $s^i|_V$, $i = 1, \ldots, n$ generate $\mathcal{F}|_V$ this means that $\varphi|_V = 0$. Since (U, φ) is equivalent to $(V, \varphi|_V)$ we conclude $\sigma = 0$ and injectivity of the map follows.

Next, assume $\mathcal{F}$ is finitely presented. By localizing on X we may assume that $\mathcal{F}$ has a presentation

$$\bigoplus\nolimits_{j=1,\ldots,m} \mathcal{O}_X \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} \mathcal{O}_X \to \mathcal{F} \to 0.$$

By Lemma 0.2 this gives an exact sequence $0 \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \to \bigoplus_{i=1,\ldots,n} \mathcal{G} \longrightarrow \bigoplus_{j=1,\ldots,m} \mathcal{G}$. Taking stalks we get an exact sequence $0 \to \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})_x \to \bigoplus_{i=1,\ldots,n} \mathcal{G}_x \longrightarrow \bigoplus_{j=1,\ldots,m} \mathcal{G}_x$ and the result follows since $\mathcal{F}_x$ sits in an exact sequence $\bigoplus_{j=1,\ldots,m} \mathcal{O}_{X,x} \longrightarrow \bigoplus_{i=1,\ldots,n} \mathcal{O}_{X,x} \to \mathcal{F}_x \to 0$ which induces the exact sequence $0 \to \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x) \to \bigoplus_{i=1,\ldots,n} \mathcal{G}_x \longrightarrow \bigoplus_{j=1,\ldots,m} \mathcal{G}_x$ which is the same as the one above. □